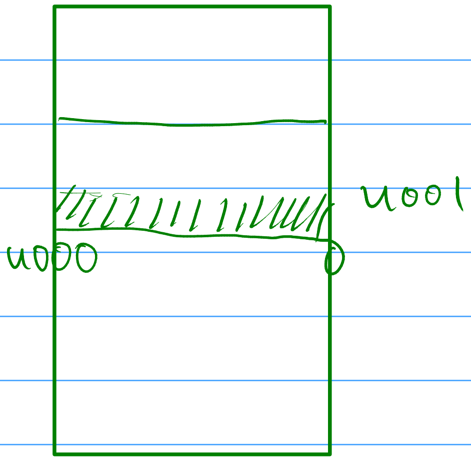
→ malloc(0)

ptr ← malloc(n) → sbrk(n)
brk(n) } → mmap - System Call

if (ptr == NULL)
error



ptr != NULL

ptr = malloc(10)

*ptr = 5;

if n=0

*ptr = 3

if (n != 0) } n=0

ptr = malloc(n x __)

*ptr=5

→

→ Semaphore to be used for process synchronization.

Sem_init (, 1, init_val);

P1

id - shmget(1, sizeof(sem_t))

ptr → shmat -

→ Sem_wait( )

→ file1

→ Sem_post( )

P2

- Sem_wait;

→ file1

→ Sem_post

O.S

PM
MM
FM
Threads
IPC
Signal Handling

} Completed.

Optimization :-

char ch, x;

ch = 5;

1 = 5

x = ch;

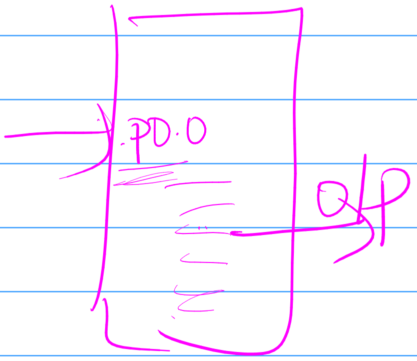
If optimization ON

```
loop
{ i = 1000000; i > 5; i++
  {

  }
};
j = i
```

Sfrs

IOPIN



volatile :-

```
while(1)
{
  if ((IOPIN0 >> 0) & 1) == 0)
  {

  }
  else
  {

  }
```

Cache register

Volatile is a keyword telling to the Compiler that not to optimize the particular variable / location.